

ACR12015

## 18-β-Glycyrrhetic acid,

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

|                                                       |                                                                                                                                                                                                                                                                    |
|-------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>产品说明:</b><br><b>Product Description:</b>           | <b>甘草亭酸</b><br><b>18-β-Glycyrrhetic acid,</b>                                                                                                                                                                                                                      |
| <b>Cat No. :</b>                                      | <b>120150000; 120150050; 120150250</b>                                                                                                                                                                                                                             |
| <b>Synonyms</b>                                       | Enoxolone                                                                                                                                                                                                                                                          |
| <b>CAS No</b>                                         | 471-53-4                                                                                                                                                                                                                                                           |
| <b>Molecular Formula</b>                              | C30 H46 O4                                                                                                                                                                                                                                                         |
| <b>Supplier</b>                                       | <b>UK entity/business name</b><br>Fisher Scientific UK<br>Bishop Meadow Road,<br>Loughborough, Leicestershire LE11 5RG, United Kingdom<br><br><b>EU entity/business name</b><br>Thermo Fisher Scientific<br>Janssen Pharmaceuticaaan 3a, 2440 Geel, Belgium        |
| <b>Emergency Telephone Number</b>                     | For information <b>US</b> call: 001-800-227-6701 / <b>Europe</b> call: +32 14 57 52 11<br>Emergency Number <b>US</b> :001-201-796-7100 / <b>Europe</b> : +32 14 57 52 99<br><b>CHEMTREC</b> Tel. No. <b>US</b> :001-800-424-9300 / <b>Europe</b> :001-703-527-3887 |
| <b>E-mail address</b>                                 | begel.sdsdesk@thermofisher.com                                                                                                                                                                                                                                     |
| <b>Recommended Use</b><br><b>Uses advised against</b> | Laboratory chemicals.<br>No Information available                                                                                                                                                                                                                  |

### SECTION 2. HAZARD IDENTIFICATION

**Physical State**  
Powder Solid

**Appearance**  
Off-white

**Odor**  
Odorless

#### Emergency Overview

The product contains no substances which at their given concentration are considered to be hazardous to health.

#### Classification of the substance or mixture

Based on available data, the classification criteria are not met

#### Label Elements

None required

#### Physical and Chemical Hazards

None identified.

#### Health Hazards

The product contains no substances which at their given concentration are considered to be hazardous to health.

#### Environmental hazards

Contains no substances known to be hazardous to the environment or not degradable in waste water treatment plants. Will likely

18- $\beta$ -Glycyrrhetic acid,

be mobile in the environment due to its water solubility. The product is water soluble, and may spread in water systems.

This product does not contain any known or suspected endocrine disruptors.

**SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS**

| Component                                                 | CAS No   | Weight % |
|-----------------------------------------------------------|----------|----------|
| Olean-12-en-29-oic acid, 3-hydroxy-11-oxo-, (3.β.,20.β.)- | 471-53-4 | > 98     |

**SECTION 4. FIRST AID MEASURES****Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur.

**Inhalation**

Remove to fresh air. Get medical attention immediately if symptoms occur.

**Ingestion**

Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.

**Most important symptoms and effects**

None reasonably foreseeable.

**Self-Protection of the First Aider**

No special precautions required.

**Notes to Physician**

Treat symptomatically.

**SECTION 5. FIRE-FIGHTING MEASURES****Suitable Extinguishing Media**

Water spray. Carbon dioxide (CO<sub>2</sub>). Dry chemical. Alcohol resistant foam.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Specific Hazards Arising from the Chemical**

Thermal decomposition can lead to release of irritating gases and vapors.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**SECTION 6. ACCIDENTAL RELEASE MEASURES****Personal Precautions**

Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation.

**Environmental Precautions**

Should not be released into the environment. See Section 12 for additional Ecological Information.

18- $\beta$ -Glycyrrhetic acid,**Methods for Containment and Clean Up**

Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

Refer to protective measures listed in Sections 8 and 13.

**SECTION 7. HANDLING AND STORAGE****Handling**

Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid contact with skin, eyes or clothing. Avoid ingestion and inhalation. Avoid dust formation.

**Storage**

Keep in a dry, cool and well-ventilated place. Keep container tightly closed.

**Specific Use(s)**

Use in laboratories

**SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION****Control Parameters****Monitoring methods**

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents. MDHS14/3 General methods for sampling and gravimetric analysis of respirable and inhalable dust

**Exposure Controls****Engineering Measures**

None under normal use conditions.

**Personal protective equipment**

**Eye Protection** Wear safety glasses with side shields (or goggles) (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material | Breakthrough time                 | Glove thickness | EU standard | Glove comments        |
|----------------|-----------------------------------|-----------------|-------------|-----------------------|
| Natural rubber | See manufacturers recommendations | -               | EN 374      | (minimum requirement) |
| Butyl rubber   |                                   |                 |             |                       |
| Nitrile rubber |                                   |                 |             |                       |
| Neoprene       |                                   |                 |             |                       |
| PVC            |                                   |                 |             |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Long sleeved clothing

**Respiratory Protection** No protective equipment is needed under normal use conditions.

**Large scale/emergency use** Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** Particle filter

## SAFETY DATA SHEET

## 18-β-Glycyrrhetic acid,

## Small scale/Laboratory use

Maintain adequate ventilation

**Recommended half mask:-** Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141

## Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice.

## Environmental exposure controls

No information available.

## SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

## Appearance

Off-white

## Physical State

Powder Solid

## Odor

Odorless

## Odor Threshold

No data available

## pH

No information available

## Melting Point/Range

292 - 295 °C / 557.6 - 563 °F

## Softening Point

No data available

## Boiling Point/Range

No information available

## Flash Point

No information available

**Method -** No information available

## Evaporation Rate

Not applicable

Solid

## Flammability (solid,gas)

No information available

## Explosion Limits

No data available

## Vapor Pressure

No data available

## Vapor Density

Not applicable

Solid

## Specific Gravity / Density

No data available

## Bulk Density

No data available

## Water Solubility

SOLUBLE IN ACETIC ACID

## Solubility in other solvents

No information available

## Partition Coefficient (n-octanol/water)

## Component

**log Pow**

Olean-12-en-29-oic acid,

4.5

3-hydroxy-11-oxo-, (3.β.,20.β.)-

## Autoignition Temperature

Not applicable

## Decomposition Temperature

No data available

## Viscosity

Not applicable

Solid

## Explosive Properties

No information available

## Oxidizing Properties

No information available

## Molecular Formula

C30 H46 O4

## Molecular Weight

470.69

## SECTION 10. STABILITY AND REACTIVITY

## Stability

Stable under normal conditions.

## Hazardous Reactions

None under normal processing.

## Hazardous Polymerization

No information available.

## Conditions to Avoid

Incompatible products.

## Materials to avoid

Strong oxidizing agents.

## Hazardous Decomposition Products

Thermal decomposition can lead to release of irritating gases and vapors. Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

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## SECTION 11. TOXICOLOGICAL INFORMATION

|                                                  |                                                                                |
|--------------------------------------------------|--------------------------------------------------------------------------------|
| <b>Product Information</b>                       | No acute toxicity information is available for this product                    |
| <b>(a) acute toxicity;</b>                       |                                                                                |
| <b>(b) skin corrosion/irritation;</b>            | No data available                                                              |
| <b>(c) serious eye damage/irritation;</b>        | No data available                                                              |
| <b>(d) respiratory or skin sensitization;</b>    |                                                                                |
| Respiratory                                      | No data available                                                              |
| Skin                                             | No data available                                                              |
| <b>(e) germ cell mutagenicity;</b>               | No data available                                                              |
| <b>(f) carcinogenicity;</b>                      | No data available<br>There are no known carcinogenic chemicals in this product |
| <b>(g) reproductive toxicity;</b>                | No data available                                                              |
| <b>(h) STOT-single exposure;</b>                 | No data available                                                              |
| <b>(i) STOT-repeated exposure;</b>               | No data available                                                              |
| Target Organs                                    | No information available.                                                      |
| <b>(j) aspiration hazard;</b>                    | Not applicable<br>Solid                                                        |
| <b>Other Adverse Effects</b>                     | The toxicological properties have not been fully investigated.                 |
| <b>Symptoms / effects,both acute and delayed</b> | No information available                                                       |

## SECTION 12. ECOLOGICAL INFORMATION

|                                      |                                                                                                                             |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b>Ecotoxicity effects</b>           | Contains no substances known to be hazardous to the environment or that are not degradable in waste water treatment plants. |
| <b>Persistence and Degradability</b> |                                                                                                                             |
| Persistence                          | Soluble in water, Persistence is unlikely, based on information available.                                                  |
| <b>Bioaccumulative Potential</b>     | Bioaccumulation is unlikely                                                                                                 |

| Component                                                                    | log Pow | Bioconcentration factor (BCF) |
|------------------------------------------------------------------------------|---------|-------------------------------|
| Olean-12-en-29-oic acid,<br>3-hydroxy-11-oxo-, (3. $\beta$ .,20. $\beta$ .)- | 4.5     | No data available             |

|                         |                                                                                            |
|-------------------------|--------------------------------------------------------------------------------------------|
| <b>Mobility in soil</b> | The product is water soluble, and may spread in water systems Will likely be mobile in the |
|-------------------------|--------------------------------------------------------------------------------------------|

18- $\beta$ -Glycyrrhetic acid,

environment due to its water solubility Highly mobile in soils

**Endocrine Disruptor Information**  
**Persistent Organic Pollutant**  
**Ozone Depletion Potential**This product does not contain any known or suspected endocrine disruptors  
This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance**SECTION 13. DISPOSAL CONSIDERATIONS****Waste from Residues/Unused Products**

Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

**Contaminated Packaging**

Empty remaining contents. Dispose of in accordance with local regulations. Do not re-use empty containers.

**Other Information**

Waste codes should be assigned by the user based on the application for which the product was used.

**SECTION 14. TRANSPORT INFORMATION****Road and Rail Transport**

Not Regulated

**IMDG/IMO**

Not regulated

**IATA**

Not regulated

**Special Precautions for User**

No special precautions required

**SECTION 15. REGULATORY INFORMATION****International Inventories**

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component                                                 | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL     |
|-----------------------------------------------------------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|----------|
| Olean-12-en-29-oic acid, 3-hydroxy-11-oxo-, (3.β.,20.β.)- | -                                                   | -                                       | X    | X     | 207-444-6 | X    | X   | -     | X    | X    | X    | KE-20783 |

**National Regulations****SECTION 16. OTHER INFORMATION****Revision Date**  
**Revision Summary**05-Apr-2024  
Not applicable.

18- $\beta$ -Glycyrrhetic acid,**Training Advice**

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

**Legend**

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**PNEC** - Predicted No Effect Concentration

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

**Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**